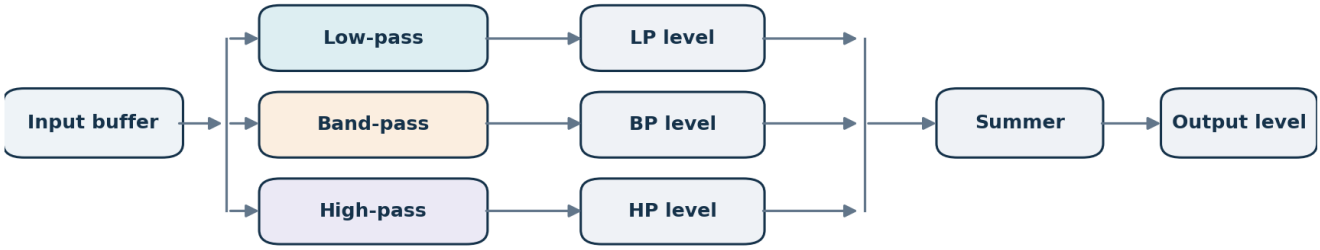
THREE-BAND ACTIVE AUDIO EQUALIZER

This report documents the intended analog filter behavior of a three-band audio equalizer using LM741 operational amplifiers. It consolidates the separate low-pass, band-pass, and high-pass calculations and uses component values from the supplied combined KiCad schematic. **All numerical response values below are theoretical calculations, not measured laboratory results.**

1. System overview

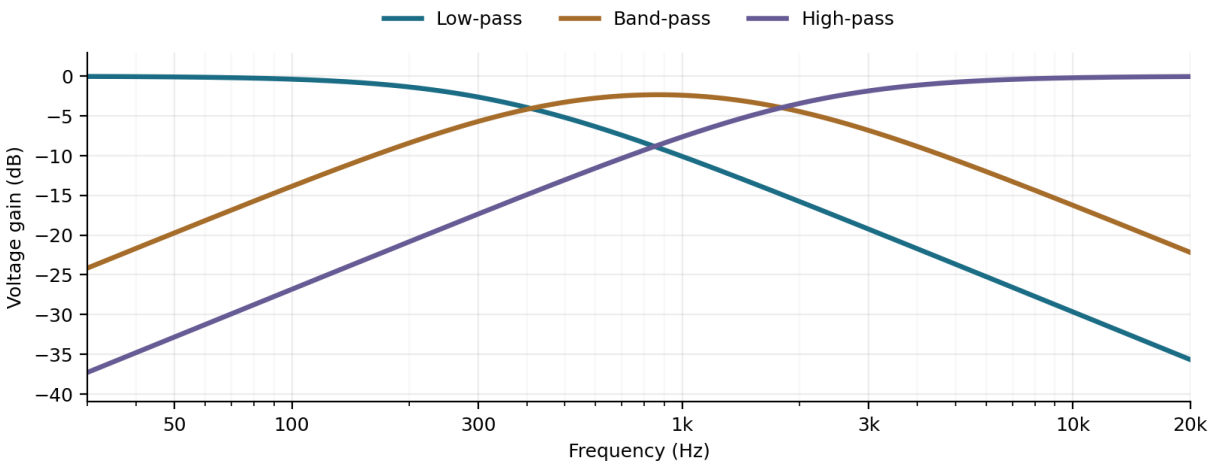
The signal is buffered, split into three frequency-selective paths, independently attenuated by potentiometers, and recombined by an output summing stage. A fourth potentiometer controls the final output level. The supplied schematic labels the op-amp supplies as **+15 V and −15 V**.



Signal architecture of the intended three-band active equalizer.

Stage	Schematic components	Frequency characteristic / role
Low-pass (LP)	R1 = 22 kΩ; R2 = 22 kΩ; C1 = 22 nF	Single-pole cutoff: 329 Hz
Band-pass (BP)	R5 = 15 kΩ; C3 = 22 nF; R6 = 15 kΩ; C4 = 6.8 nF	RC corners: 482 Hz, 1.56 kHz; peak near 867 Hz
High-pass (HP)	R3 = 2.2 kΩ; C2 = 33 nF; R4 = 2.2 kΩ	Single-pole cutoff: 2.19 kHz
Mixing and output	POT1–POT4, R7–R12, LM741 summer	Independent level setting and recombination

Theoretical individual-stage frequency response



2. Low-pass filter

Purpose. Preserve lower-frequency audio while attenuating frequencies above the cutoff. The individual calculation uses an inverting op-amp with a resistor at the input and a parallel resistor–capacitor feedback network.

Impedance and transfer-function derivation

Taking $s = j\omega$, and using $Z_C = 1/(sC)$, the input and feedback impedances are:

$$Z_{\text{in}} = R_1$$

$$Z_f = R_2 \parallel \frac{1}{sC_1} = \frac{R_2}{1 + sR_2C_1}$$

For the *intended inverting* topology with ideal negative feedback, $H(s) = -Z_f/Z_{\text{in}}$:

$$H_{\text{LP}}(s) = -\frac{R_2/R_1}{1 + sR_2C_1}$$

The magnitude decreases as frequency increases:

$$|H_{\text{LP}}(j\omega)| = \frac{R_2/R_1}{\sqrt{1 + (\omega R_2C_1)^2}}$$

Component selection and cutoff frequency

Component	Chosen value	Function
R1	22 kΩ	Input resistance
R2	22 kΩ	Feedback resistance; sets DC gain and cutoff
C1	0.022 µF = 22 nF	Feedback capacitance

The DC gain is $-R_2/R_1 = -1$ V/V (magnitude 1 V/V, or 0 dB). At the −3 dB cutoff:

$$f_{c,\text{LP}} = \frac{1}{2\pi R_2C_1}$$

$$f_{c,\text{LP}} = \frac{1}{2\pi(22 \times 10^3)(22 \times 10^{-9})} = 328.83 \text{ Hz}$$

The original low-pass calculation targeted approximately **320 Hz**, with a **22 kΩ** standard resistor and **22 nF** capacitor. These chosen parts produce a nominal cutoff of **329 Hz**, close to that target.

Expected behavior

At low frequency, $H_{\text{LP}} \to -1$; at the cutoff, the magnitude is $1/\sqrt{2} \approx 0.707$ V/V; and far above the cutoff the magnitude falls at approximately **20 dB/decade**. The negative sign indicates a nominal 180° signal inversion within the passband.

3. Band-pass filter

Purpose. Emphasize a middle-frequency band by combining a capacitive high-pass input path with a capacitive low-pass feedback path around an inverting op-amp. The separate band-pass notes derive the same topology shown in the component list below.

Impedance and transfer-function derivation

The series input branch consists of R_5 and C_3. The feedback branch consists of R_6 in parallel with C_4:

$$Z_{\text{in}} = R_5 + \frac{1}{sC_3} = \frac{1 + sR_5C_3}{sC_3}$$

$$Z_f = R_6 \parallel \frac{1}{sC_4} = \frac{R_6}{1 + sR_6C_4}$$

The ideal inverting transfer function is therefore:

$$H_{\text{BP}}(s) = -\frac{sR_6C_3}{(1 + sR_5C_3)(1 + sR_6C_4)}$$

At very low frequency, the series input capacitor blocks the signal. At very high frequency, the feedback capacitor reduces the feedback impedance. The combination passes a middle range.

Selected parts and characteristic frequencies

Component	Value	Function
R5	15 kΩ	Series input resistor
C3	0.022 µF = 22 nF	Series input capacitor
R6	15 kΩ	Feedback resistor
C4	6.8 nF	Feedback capacitor

The two *individual RC corner frequencies* in the transfer-function denominator are:

$$f_{\text{low,RC}} = \frac{1}{2\pi R_5C_3} = \frac{1}{2\pi(15 \times 10^3)(22 \times 10^{-9})} = 482.29 \text{ Hz}$$

$$f_{\text{high,RC}} = \frac{1}{2\pi R_6C_4} = \frac{1}{2\pi(15 \times 10^3)(6.8 \times 10^{-9})} = 1560.34 \text{ Hz}$$

The peak-response frequency for this ideal first-order high-pass / low-pass combination is the geometric mean:

$$f_{\text{peak}} = \sqrt{f_{\text{low,RC}} f_{\text{high,RC}}} = 867.49 \text{ Hz}$$

Its theoretical peak magnitude is:

$$|H_{\text{BP}}|_{\text{peak}} = \frac{R_6C_3}{R_5C_3 + R_6C_4} = 0.764 \text{ V/V}$$

Interpretation. The **482 Hz** and **1.56 kHz** values describe separate RC corners; they are not the −3 dB limits of the *combined* band-pass response. The composite response has approximately **319 Hz** and **2.36 kHz** half-power frequencies relative to its peak, owing to the overlap of the two gradual slopes.

4. High-pass filter

Purpose. Attenuate low frequencies while retaining higher-frequency content. In the original high-pass notes, a series resistor–capacitor input drives an inverting stage with resistive feedback.

Transfer function and component selection

For the integrated design, the input branch is R_3 and C_2 in series, while the feedback branch is R_4:

$$Z_{\text{in}} = R_3 + \frac{1}{sC_2} = \frac{1 + sR_3C_2}{sC_2}, \qquad Z_f = R_4$$

$$H_{\text{HP}}(s) = -\frac{sR_4C_2}{1 + sR_3C_2}$$

The high-frequency passband gain approaches $-R_4/R_3 = -1$ V/V. The cutoff frequency is:

$$f_{c,\text{HP}} = \frac{1}{2\pi R_3C_2}$$

$$f_{c,\text{HP}} = \frac{1}{2\pi(2.2 \times 10^3)(33 \times 10^{-9})} = 2192.22 \text{ Hz}$$

Component	Value	Function
R3	2.2 kΩ	Series input resistor
C2	0.033 µF = 33 nF	Series input capacitor
R4	2.2 kΩ	Feedback resistor

Below **2.19 kHz**, the magnitude decreases toward zero as the signal frequency falls. Above the cutoff, it approaches unity (0 dB). This is a nominal first-order filter with approximately **20 dB/decade** low-frequency rise.

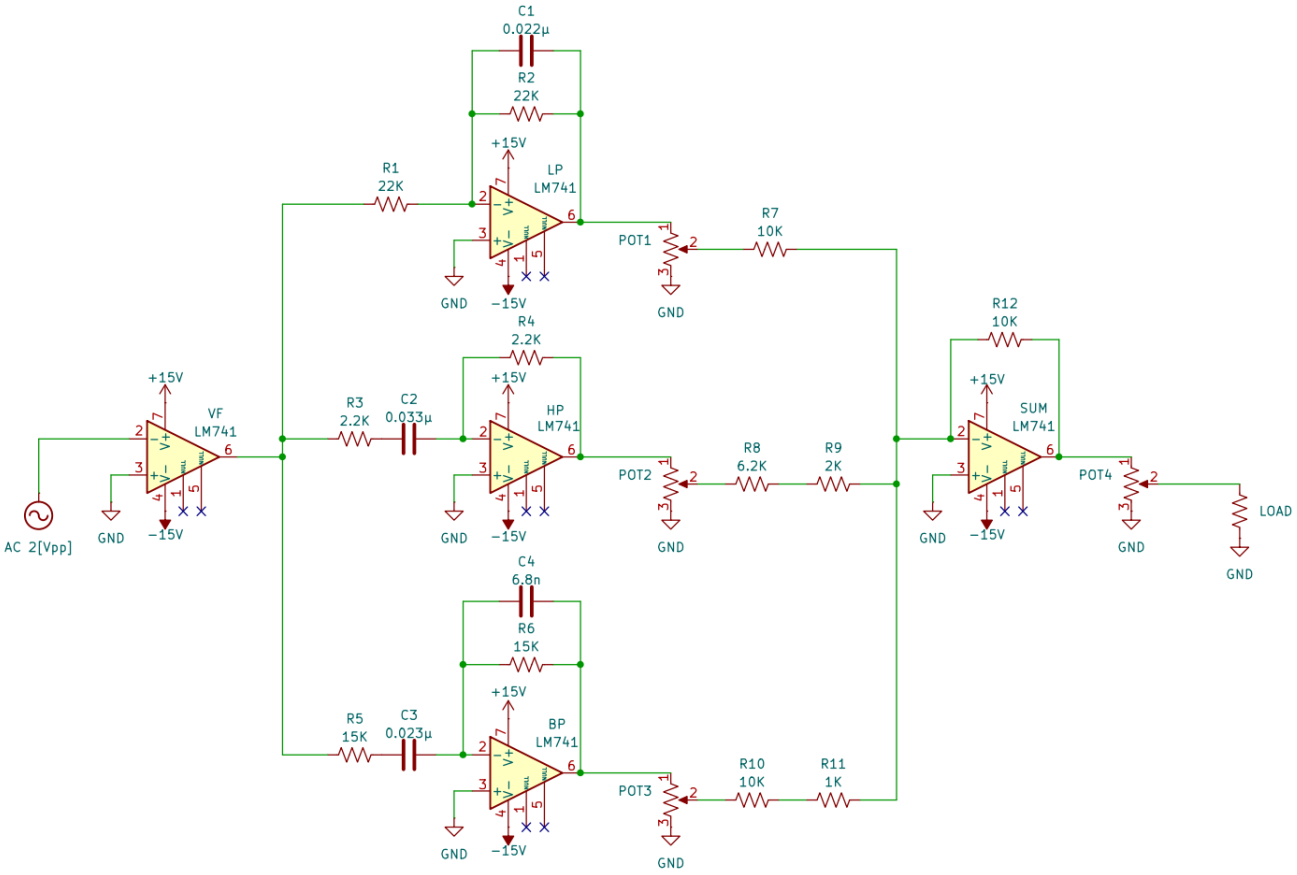
5. Level controls and summing stage

Each filter output feeds a potentiometer (POT1, POT2, and POT3). The potentiometer wipers pass through their respective input resistors to a common op-amp summer. For a correctly wired **inverting summing amplifier**, the ideal relationship would be:

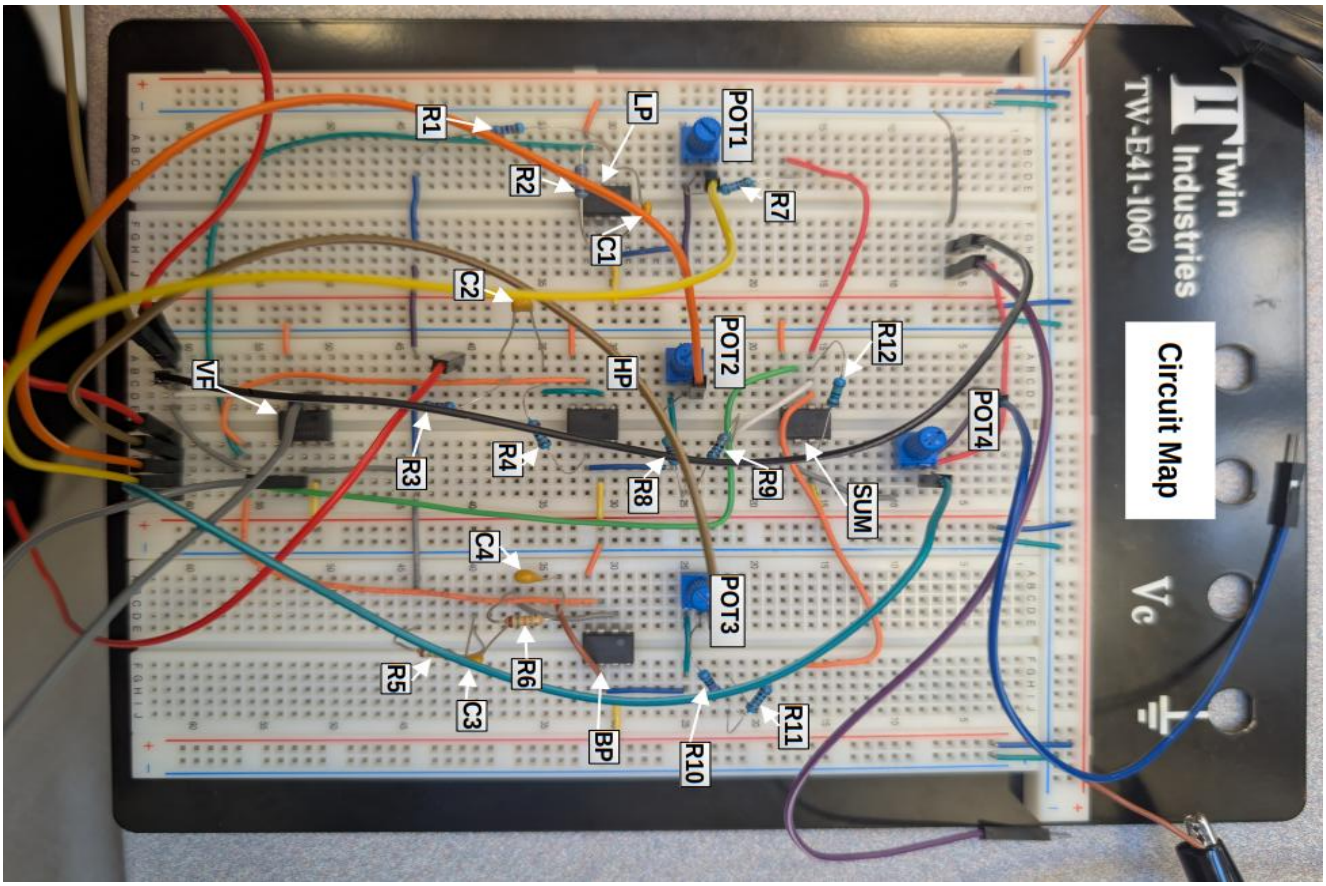
$$V_{\text{SUM}} = -R_{12}\left(\frac{\alpha_{\text{LP}}V_{\text{LP}}}{R_7} + \frac{\alpha_{\text{HP}}V_{\text{HP}}}{R_8 + R_9} + \frac{\alpha_{\text{BP}}V_{\text{BP}}}{R_{10} + R_{11}}\right)$$

Here α denotes each control's ideal unloaded wiper fraction (0–1); actual gain also depends on loading. With the shown resistor values, the nominal contribution magnitudes are **1.00**× (LP: 10 kΩ), **1.22**× (HP: 8.2 kΩ), and **0.91**× (BP: 11 kΩ) before wiper attenuation. R_{12} is **10 kΩ**. POT4 acts as the final output level control; its value and the load value are not specified in the supplied drawing.

6. Source schematic and Circuit Buildout



Circuit Map:



7. Simulated Waveforms

